

Evaluation Report & Next-Step Execution Roadmap

Intern Name: VEDAKSHARI

Assigned Construct: Visual Processing (Gv)

Proposed Task: Mystery Map Builder

Evaluation Status: Proceed with Significant Enhancements for Research-Grade MVP

1. Executive Summary

The proposed Mystery Map Builder is a strong and engaging assessment concept that aligns well with Visual Processing (Gv) in CHC Theory. The task successfully incorporates mental rotation, spatial relations, visualization, visual closure, spatial scanning, and visual memory through a puzzle-based map completion activity. However, the current version resembles a static puzzle test rather than a scalable AI-powered cognitive assessment. To support startup objectives, the task must evolve into a procedurally generated, adaptive, behaviorally logged assessment engine capable of producing thousands of unique map challenges and capturing rich cognitive analytics.

2. Alignment with MentiScope's Objectives

- Alignment with CHC-Gv Theory: 9/10
- Scientific Validity: 8.5/10
- Innovation: 8/10
- Adaptive Potential: 9/10
- AI Readiness: 8.5/10
- MVP Feasibility: 9/10
- Overall Evaluation: 8.8/10 – Proceed with Enhancements

3. Scientific Assessment

Primary Gv Components Measured:

- Visualization (Vz)
- Spatial Relations / Mental Rotation (SR)
- Visual Closure (CS)
- Flexibility of Closure (CF)
- Spatial Scanning (SS)
- Visual Memory (MV)

Potential Construct Contamination:

- Processing Speed (Gs)
- Fine Motor Skills
- Attention/Vigilance
- Puzzle familiarity

The assessment should focus on visual-spatial reasoning rather than drawing accuracy or gaming skills.

4. Recommended Task Architecture

Module 1 – Piece Identification

Measures visual closure and flexibility of closure.

Module 2 – Mental Rotation Challenge

Measures spatial relations and mental rotation.

Module 3 – Spatial Placement

Measures visualization and spatial reasoning.

Module 4 – Distractor Elimination

Measures visual discrimination and scanning.

Module 5 – Adaptive Mystery Maps

Map size, rotations, distractors, and complexity automatically adapt to performance.

5. Recommended Scoring Model

- Piece Selection Accuracy – 25%
- Mental Rotation Accuracy – 25%
- Spatial Placement Accuracy – 20%
- Visual Strategy & Error Control – 15%
- Efficiency & Completion Time – 15%

6. AI Analytics to Capture

- Rotation attempts
- Number of corrections
- Distractor selection rate
- Search path patterns
- Time per placement
- Error clusters
- Strategy shifts
- Visual scanning efficiency
- Completion confidence index
- Learning curve

7. MVP Deliverables (Current Phase)

- ✓ Procedural map generator
- ✓ Puzzle-piece generation engine
- ✓ Rotation engine
- ✓ Adaptive difficulty framework
- ✓ Event logger

- ✓ Analytics dashboard
- ✓ Documentation
- ✓ Pilot dataset

8. Assessment SDK Compliance Requirements

Session Data: student_id, session_id, construct, task_id, module_version, difficulty_level, start_time, end_time, timestamp.

Event Data: event_id, student_id, session_id, construct, task_id, item_id, event_type, response, correct, reaction_time_ms, error_type, difficulty_level, timestamp.

Score Output: raw_score, normalized_score, percentile, sub_scores, confidence_score.

Analytics Output: behavioral_metrics and recommendations.

9. Next Sprint Execution Plan (2 Weeks)

1. Review CHC Visual Processing literature.
2. Define sub-ability mapping for each task component.
3. Build procedural map generator.
4. Develop rotation and placement engine.
5. Create distractor-generation algorithm.
6. Implement adaptive difficulty rules.
7. Implement SDK-compliant event logging.
8. Build one fully functional MVP module.

10. Definition of Done (DoD)

Scientific:

- ✓ All Gv subcomponents clearly mapped.
- ✓ Construct contamination minimized.
- ✓ Difficulty calibrated.

Technical:

- ✓ Procedural generation implemented.
- ✓ Adaptive engine functional.
- ✓ Analytics dashboard available.
- ✓ SDK export available.

Validation:

- ✓ Internal consistency ≥ 0.80 .
- ✓ Pilot study with 100 students.
- ✓ Convergent validation with established Gv measures.
- ✓ Reliability evidence documented.

Guidance to the Intern

Your responsibility is not to build a puzzle game. Your responsibility is to build a scientifically valid Visual Processing assessment engine. Focus on procedural generation, adaptive complexity, behavioral analytics, and CHC-aligned measurement. Every student should receive a unique map while the platform captures standardized data for future AI-based cognitive profiling.

